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ABSTRACT

Ammonia-fueled solid oxide fuel cell (SOFC) systems with anode off-gas recirculation (AOGR) offer high efficiency for marine power generation. In particular, ejector-aided AOGR can further improve electrical efficiency as it requires no external power. However, system performance is strongly coupled to operating conditions and the AOGR ratio, while previous studies have generally treated the AOGR ratio as a freely adjustable design input rather than a physically constrained outcome and examined its effect along a single operating axis. This study addresses these limitations by identifying the physically attainable maximum AOGR ratio and quantifying its interactions with fuel utilization, current density, and stack inlet temperature. The analysis was conducted using an integrated thermodynamic model with validated component models, implemented within a commercial thermodynamic simulator (H-Twin[®]). Results show that increasing fuel utilization from 40% to 80% raises the maximum AOGR ratio from 65.4% to 66.9%, whereas increasing current density from 0.1 to 0.5 A/cm² lowers it from 67.3% to 62.0%. Elevated stack inlet temperature further lowers this limit by degrading ejector performance. The AOGR ratio consistently improves efficiency while reducing power density, with the largest efficiency gains occurring at low fuel utilization. In addition, the optimal stack operating temperature shifts downward as the AOGR ratio increases. At high current density, growing balance-of-plant parasitic power produces a non-monotonic efficiency trend, yielding an optimal AOGR ratio below its maximum allowable value. These findings demonstrate that the AOGR ratio, operating conditions, and system constraints must be co-optimized for effective ammonia-fueled SOFC system design.

1. Introduction

The maritime industry is currently undergoing a major transition toward decarbonization, driven by tightening international regulations on greenhouse gas emissions and the long-term goal of achieving net-zero shipping [1, 2]. As conventional marine fuels such as heavy fuel oil and marine diesel oil are no longer compatible with these targets, the search for clean and energy-dense alternative fuels has become a central topic in marine power generation research [3]. Among the candidates proposed thus far, ammonia has emerged as one of the most prominent options because it contains no carbon and can therefore be utilized without any direct CO₂ emission [4]. Compared with hydrogen, which has often been regarded as the ultimate clean fuel but suffers from cryogenic storage requirements and low volumetric energy density, ammonia can be stored and transported as a pressurized liquid under significantly milder conditions,

making it far more practical for shipboard storage where space and weight constraints are critical [5, 6]. Furthermore, an extensive global infrastructure for ammonia production, transportation, and bunkering has already been established for fertilizer use, providing a uniquely favorable starting point for its near-term deployment as a marine fuel [7].

To convert ammonia fuel into electrical power onboard, a power conversion device that combines high efficiency with operational compatibility with ammonia is required. Internal combustion engines are an option currently under active development, but their combustion processes inherently produce NO_x and N₂O emissions that necessitate additional flue-gas treatment systems [8]. Fuel cells, in contrast, electrochemically convert fuel into electricity without combustion, thereby offering both higher efficiency and substantially cleaner emissions [9]. Among various fuel cell types investigated for ammonia operation, solid oxide fuel cell (SOFC) is particularly well-suited because its high operating

*Corresponding author.

 jongsup.hong@yonsei.ac.kr (J. Hong)

ORCID(s):

36 temperature, typically between 600 and 900 °C, simulta- 69
37 neously provides high thermodynamic efficiency and the 70
38 thermal driving force necessary to decompose ammonia into 71
39 hydrogen and nitrogen [10, 11]. For these reasons, ammonia- 72
40 fueled SOFCs have rapidly emerged as a strong candidate for 73
41 next-generation marine power generation systems pursuing 74
42 zero-emission shipping. 75

43 Reflecting such potential, extensive research efforts have 76
44 been conducted on ammonia-fueled SOFC systems, ranging 77
45 from fundamental electrochemical performance evaluation 78
46 to system-level thermodynamic analysis [12, 13, 14, 15, 16]. 79
47 Wojcik et al. [12] experimentally evaluated ammonia as 80
48 a fuel for SOFCs and demonstrated that ammonia-fed op- 81
49 eration can achieve performance comparable to hydrogen- 82
50 fed operation when ammonia decomposition is effectively 83
51 promoted over suitable catalyst materials. Cinti et al. [13] 84
52 combined ammonia-fed SOFC stack testing with system- 85
53 level analysis and showed that SOFC operation with ammo- 86
54 nia is technically feasible. Ishak et al. [14] performed ther- 87
55 modynamic analysis of ammonia-fed SOFCs and showed 88
56 that ammonia can serve as an effective hydrogen carrier for 89
57 SOFC operation, with system performance strongly affected 90
58 by operating conditions such as stack temperature and fuel 91
59 utilization. Lee et al. [15] thermodynamically compared 92
60 internal and external ammonia decomposition strategies for 93
61 ammonia-fueled SOFC systems, showing that thermal inte- 94
62 gration methods and system configurations can greatly affect 95
63 system performance. Ishak et al. [16] conducted energy and 96
64 exergy analyses of direct ammonia SOFC system integrated 97
65 with a gas turbine cycle, showing that the hybrid configu- 98
66 ration can effectively utilize SOFC waste heat for combined 99
67 heat and power generation, improving overall system perfor-100
68 mance. 101

Previous studies have demonstrated that the performance of SOFC systems is strongly influenced by a wide range of design and operating parameters. A promising design strategy to enhance the electrical efficiency of SOFC systems is anode off-gas recirculation (AOGR). AOGR improves system electrical efficiency by decreasing the amount of fresh fuel consumption at a fixed SOFC stack fuel utilization condition, thereby increasing the net system fuel utilization [17]. To thoroughly analyze the effect of AOGR on ammonia-fueled SOFC systems, various studies have been conducted. Selvam et al. [18] thermodynamically analyzed a full AOGR ammonia-fueled SOFC system achieving 100% system fuel utilization, showing that the full AOGR strategy improves energy and exergy efficiencies by 12.17 and 11.39 percentage points, respectively, over a conventional system. Jeong et al, Bakała et al. [19, 20] compared the performance of SOFC systems fueled with hydrogen, natural gas and ammonia, under both with and without AOGR. Both studies revealed that AOGR generally tends to improve system electrical efficiency, regardless of fuel types. Quach et al. [21] studied the effect of water removal at the anode off-gas via condensation in ammonia-fueled SOFC system, and further improved system performance via integration of an internal combustion engine. The efficiency of AOGR system with water removal was 12.7 percentage point higher than that of a standalone system, and was only 0.1 percentage point lower than that of an AOGR system utilizing both water removal and internal combustion engine. Sun et al. [22] conducted a parametric study to analyze how operating conditions such as current density, AOGR ratio, etc. affect the system performance of an ammonia-fueled SOFC system. Oh et al. [23] also conducted a parametric study to find the optimal operating condition of an ammonia

fueled-SOFC system undergoing AOGR, integrated with an organic Rankine cycle.

However, while previous studies focus on proposing high-efficiency system architectures, they tend to lack practical thermodynamic considerations in system evaluation regarding AOGR. First, prior studies consider AOGR ratio as an independently adjustable operating variable, whereas in practice an upper bound may exist due to physical constraints such as pressure drops in BoPs [18, 21, 22, 23]. Second, the influence of AOGR ratio has typically been examined along a single operating axis at a fixed fuel utilization or current density, leaving the coupling between AOGR ratio and other operating variables unexplored [21, 22, 23]. Third, prior studies either do not account for or assume a designated value for pressure drop effects at heat exchangers [18, 20, 21, 22, 23]. Furthermore, in systems utilizing an ejector for AOGR, the ejector model is overly simplified as a component which mixes two streams and pressurizes the entrained flow by a fixed pressure ratio [22].

To address such limitations, the present study pursues the following research objectives with the following novelties:

- The physically attainable maximum AOGR ratio is identified as an endogenous outcome of component level hydrodynamics under varying stack inlet temperature, fuel utilization, and current density, rather than being prescribed as an external design input.
- A two-dimensional parametric study is conducted in a system level to reveal how the AOGR ratio couples with each operating variables, thereby analyzing the mechanisms by which the AOGR ratio influences overall system performance under realistic operating conditions.

- A high-fidelity ammonia-fueled SOFC system model is developed, in which pressure drops across balance-of-plant components are evaluated through empirical correlation based physical models accounting for design variables, and the AOGR is described by a reliable one-dimensional ejector model that captures the entrainment-pressure-lift behavior.

2. Thermodynamic Component Modeling

Methodology

To study the performance of ammonia-fueled SOFC systems under various operating conditions, thermodynamic systems were computationally modelled and analyzed via numerical simulation. Numerical simulations were conducted via a commercial solid oxide cell system simulator, H-Twin® (H-Cube Solutions). The thermodynamic systems consisted of system components, fluid streams, and an integrated system. An integrated system was assembled by a set of system components which are connected by fluid streams represented by thermodynamic properties including mass flow rate ($\dot{m}$), temperature (T), pressure (p) and chemical composition (y_i). The system components of an ammonia-fueled SOFC system include SOFC stack, heat exchanger, ammonia reformer, ejector, blower, etc. Thermodynamic properties are calculated for each stream via a steady-state numerical solver, where each system component calculates the thermodynamic properties of its outlet stream as the output, given the inlet stream and its design configurations as the input. Numerical values of thermodynamic properties were acquired from the following sources: thermodynamic properties of water [24], specific heat capacities of gases [25], thermal conductivity and dynamic viscosity of gases [26], and enthalpy of formation of gases [27]. Design configuration of system components include geometrical

design variables and operational parameters. For example, in the SOFC stack model, geometrical design variables such as cell dimensions and operational parameters such as current density are determined as design configurations. These are combined with the inlet-stream thermodynamic properties to solve electrochemical and mass/energy balance equations that determine the outlet stream properties, cell voltage and power output. Throughout the study, key assumptions applied are as follows:

- For chemical species, only H_2 , H_2O , O_2 , N_2 , NH_3 are considered.
- For the equation of state, only H_2O is modeled as a real gas while other gases are modeled as ideal gases.
- In energy conservation, kinetic and potential energy are neglected.
- All system components assume an adiabatic condition between the component and the environment.
- Radiative heat transfer is neglected.
- Steady-state operation is assumed.
- Pressure drop does not occur in pipes when connecting components.
- Effect of stacking SOFC cells is neglected.

2.1. SOFC Stack

A one-dimensional model of an ammonia-fueled SOFC stack developed by Lee [28] was employed to calculate the voltage, power generation and thermodynamic properties of its outlet streams. The model extends the one-dimensional SOFC stack model of Min et al. [29] by incorporating ammonia decomposition kinetics at the anode. It considers discretized sections of a SOFC stack in the direction of fuel flow and solves complete analytical expressions of

the electrochemical and thermodynamic relations for each section to derive its output parameters. Given the input parameters, the internal electrochemical reaction together with the ammonia decomposition reaction is solved, and the local value of voltage of each section is calculated by the following equations.

$$V = E_{rev} - \eta_{act,a} - \eta_{act,c} - \eta_{ohm} - \eta_{conc,a} - \eta_{conc,c} \quad (1)$$

$$E_{rev} = -\frac{\Delta G^\circ}{2F} + \frac{RT}{2F} \ln \left(\frac{p_{H_2} p_{O_2}^{\frac{1}{2}}}{p_{H_2O}} \right) \quad (2)$$

Detailed expressions and parameters of the overpotentials are provided in [29], and those of the ammonia decomposition kinetics in [28]. The baseline model was validated against experimental data for methane operation [29], and the ammonia extension was further validated for two fuel conditions—pure ammonia and a 1:3 N_2/H_2 mixture representing fully decomposed ammonia [28]. Since ammonia is decomposed upstream in the fuel reformer in this study, the anode feed of the present system is nearly free of ammonia and in-stack decomposition is correspondingly small; the kinetic submodel nevertheless allows a single stack model, validated from pure ammonia to fully decomposed feed, to be applied consistently over the operating conditions examined here. The calculated values of voltage at each section are averaged to derive the final voltage of the stack. Based on the value of voltage, design configurations and operational parameters of the stack, gross power generation ($\dot{W}_{gross}$) is calculated by the following equation,

$$\dot{W}_{gross} = j A_{cell} N_{cell} V_{SOFC} \quad (3)$$

where j is the current density, A_{cell} is cell effective area, N_{cell} is the number of stacked cells, V_{SOFC} is the unit cell voltage. Along with the calculation of voltage, the heat

generated from electrochemical reaction is utilized for outlet gases post – heating, satisfying the mass, energy and heat balance in each section and the overall stack.

2.2. Balance of Plants (BoPs)

2.2.1. Heat Exchanger

A shell and tube heat exchanger model developed by Park et al. [30] was employed to simulate heat transfer from post – combustion gas to fresh air. Fresh air flows into the cold side (tube side) of the heat exchanger as cold fluid, and post combustion gas enters the hot side (shell side) as the hot fluid. The model takes geometrical parameters such as tube length, shell diameter, number of baffles and thermodynamic properties of inlet gases as the input to calculate output values such as pressure drops, heat transfer coefficient, outlet gas temperature. Such output parameters are calculated from input parameters by the Bell – Delaware method.

2.2.2. Reformer

An ammonia reformer model developed by Lee et al. [15] was exploited to simulate ammonia decomposition using heat from the high temperature stack off-gas. The reformer model shares the same dimensional properties as the shell and tube heat exchanger model while the tube is packed with cylindrical pellets of Ni catalyst. Pure ammonia, or a mixture of ammonia and anode off-gas flows into the cold side as the cold fluid and stack off-gas enters the hot side as the hot fluid. It is assumed that the composition of outlet gas of the cold side reaches chemical equilibrium, as ammonia is fully decomposed above 600 °C in Ni-based catalysts [31, 32, 33].

2.2.3. Ejector

An ejector model developed by Park et al. [34] was employed for AOGR. The model has been validated in the same study by comparing simulation results with data from a prior study accounting ejector performance in a SOFC AOGR

system [35]. Performance parameters of the validated ejector model are provided in Supplementary Table S1. The ejector model allows AOGR with zero parasitic power by ‘inhaling’ the low-pressure anode off-gas as entrained flow, using high-pressure ammonia as primary flow. As a result, the ejector discharges a mixture fluid as a fuel for SOFC stack. The ejector model accounts for compressible flow effects by assuming critical mode operation, where both the primary and secondary stream undergo choking. Input parameters are the thermodynamic properties of both flows entering the ejector, and the model calculates a suitable geometry for mass flow rates of the inlet flows as the output, together with the thermodynamic properties of the mixed stream.

2.2.4. Others

Other BoPs composing the system include blower, splitter, combustor, etc. Under a similar underlying mechanism, the BoPs calculate the thermodynamic properties of its outlet stream by calculating mass – energy balance relations together with governing equations depending on the function of the BoP. Details of each model can be found in reference [30].

3. System Configuration

3.1. System Layout Design

When designing a SOFC system, a suitable system integration method must be adopted to satisfy thermal conditions such as the feasibility to supply sufficient heat to SOFC inlet flows. Accordingly, two system layouts were evaluated in this study: configurations with and without AOGR. The process flow diagrams of both layouts are presented in Fig. 1.

3.1.1. Recirculation System

The process flow diagram of the recirculation system layout is shown in Fig. 1a. This layout was obtained by

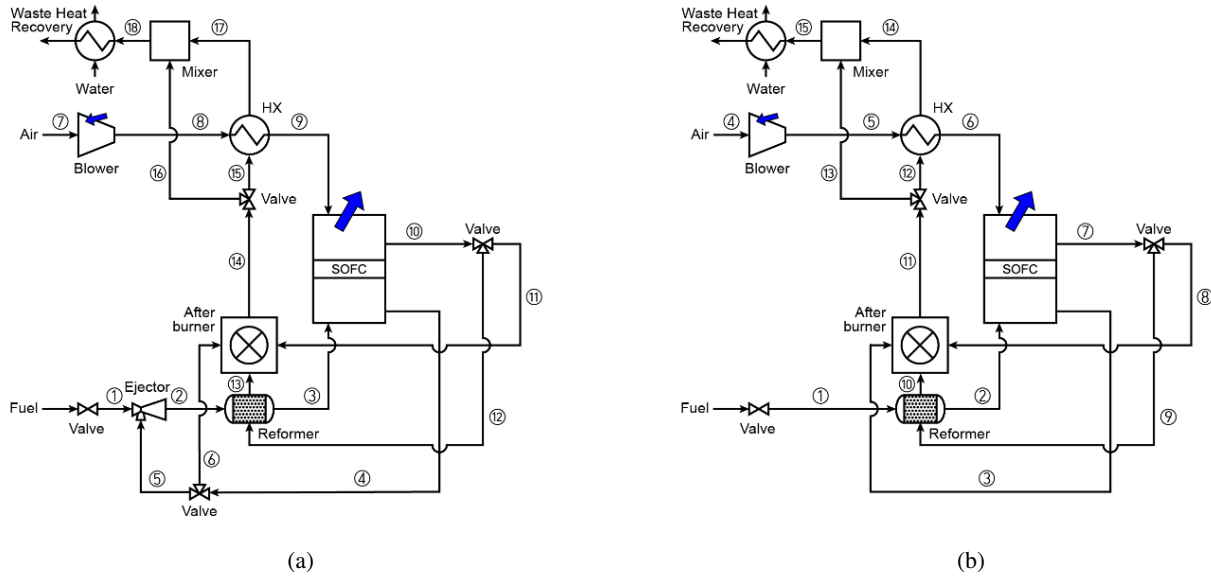


Figure 1: System layouts: (a) recirculation system; (b) non-recirculation system.

referencing Ejector CG layout proposed by Park [34] as a baseline and modifying it to meet the objectives of this study. The three limitations of the baseline layout under the operating conditions considered in this study, together with the resolution applied to each of them, are described in the remainder of this section.

Ejector CG is a layout where a portion of anode off-gas is recirculated as fuel, and the rest is utilized as a hot stream of the reformer. Air and fuel are pressurized by a blower or compressor to compensate for pressure drop effects and heated in a heat exchanger or reformer before entering the SOFC stack. By adjusting the split ratio of the splitter (three-way valve), the system can achieve operational flexibility via control of AOCR rate. The non recirculated part of anode off-gas flows into the hot side of the reformer to provide heat to the recirculated fuel. In addition, the remaining heat of both anode and cathode off-gas can be efficiently utilized by a sequence of raising the temperature of off-gas mixture through combustion, utilizing the heat at the air heat exchanger, and finally extracting waste heat from the waste heat recovery system. However, the prior study

of Ejector CG layout was conducted for natural gas fueled SOFC system, and operational parameters were not varied. Since this study analyzes ammonia fueled SOFC systems, and operational parameters are varied at a wide range, ejector CG layout has the following limitations.

1. Depending on system operating conditions, the system may not be thermodynamically feasible.
2. For a given design configuration of a heat exchanger, the required mass flow rate of hot stream at each heat exchanger differs for every single operating condition.
3. Unnecessary parasitic power is induced at the fuel compressor.

An exemplary operating condition when stack fuel utilization = 70%, anode inlet $T = 600\text{ }^{\circ}\text{C}$, current density = 0.2 A/cm^2 , AOCR ratio = 50% – which is introduced afterwards as the base case – is a case where limitation 1 may occur. The cracking of ammonia is a highly endothermic reaction, hence a large amount of heat duty is required at the reformer. However, the anode off-gas does not contain sufficient thermal energy to supply heat to the cold stream

of the reformer due to its relatively low mass flow rate. Therefore, in such cases, the thermal energy rich cathode off-gas could be utilized alternatively as the heat source to supply sufficient heat to the cold stream, while satisfying the 2nd law of thermodynamics. A T-Q diagram analysis of each cases – utilizing cathode off-gas as the reformer heat source and anode off-gas as the reformer heat source – is displayed in [Supplementary Fig. S1](#) to quantitatively verify the need of utilizing cathode off-gas as the heat source of the fuel reformer.

To resolve limitation 2, the heat duty value at both the heat exchanger and reformer must be adjusted to provide a suitable amount of heat to the cold side fluid. However, the heat duty required at each point differs for every different operating condition of the system. Since the heat exchanger model and reformer model calculates heat duty and outlet temperature based on design configuration and inlet thermodynamic properties, heat duty can be adjusted by differing the heat exchanger's geometrical parameters or differing the mass flow rate entering the hot side. Mass flow rate of the hot side fluid greatly affects the amount of heat transfer between the tube and hot side fluid, as a greater mass flow rate substantially increases turbulence in the hot side flow, improving heat transfer performance [36]. The design configurations of a shell and tube heat exchanger is a unique parameter which cannot be adjusted flexibly in real engineering applications; therefore, hot side inlet mass flow rate is a readily available variable that can be varied. Hot side inlet mass flow rate can be adjusted by adding a splitter upstream of the heat exchanger and reformer's hot side flow. The splitter is given a designated split ratio to branch off a suitable mass flow rate of hot gas to the heat exchanger and reformer, so that the cold outlet stream can reach a desired temperature. In the case of heat exchanger,

mass flow rate is branched off to provide a suitable amount of heat for air heating, and in the case of reformer, mass flow rate is branched off to provide a suitable amount of heat for ammonia decomposition and fuel heating.

Limitation 3 occurs due to a difference in the target environment and fuel of the system. The study conducted by Park [34] analyzed a natural gas fueled SOFC system for residential heat and power generation, thereby assuming that natural gas is fed into the system at 1 atm. To induce proper operation at the ejector and compensate for pressure drop effects at the fuel reformer, natural gas is pressurized by the compressor at a high pressure of over 5 atm in the corresponding study. Such operation causes a significant increase in parasitic power due to power consumption of the fuel compressor. However, this study considers ammonia as a fuel for marine power generation, where ammonia is conventionally stored at a high pressure of 18 bar for fuel [37]. Therefore, given the sufficiently high inlet pressure of pure ammonia, the fuel compressor is omitted, and a throttling valve is installed instead to meet the required pressure boundary conditions across operating points.

3.1.2. Non-Recirculation System

The process flow diagram of the non – recirculation system layout is shown in Fig. 1b. In a case where the AOGR ratio is 0, the ejector does not play any role in the system as no part of the anode off-gas is fed back, and pure ammonia fuel from the system inlet bypasses through the ejector. To evaluate non recirculation system performances under unified conditions, an identical system layout was utilized, but the ejector was omitted from the process flow diagram together with the recirculating splitter. Since the splitter is omitted, all parts of the anode off-gas is directly fed to the combustor to provide heat to the air heat exchanger after combustion.

3.2. Input Parameters & Boundary Conditions

To compare system performances at different operating conditions, a set of common input parameters must be established to solely analyze the effect of changing operating parameters. The input parameters and boundary conditions imposed in this study are summarized in Table 1. The ambient conditions are set as 25°C and 1 atm, thereby setting the inlet condition of air as the same temperature and pressure. Due to the ambient condition, a pressure boundary condition is set at the system sink to be 1 atm, so that neither fluid backflow occurs at the sink nor any unnecessary parasitic power is consumed at the air blower. In addition, system outlet temperature is set to dew point, as a waste heat recovery system recuperates heat from the exhaust gas until it reaches dew point. At the fuel inlet side, 100% ammonia is fed as a saturated vapor at 18 bar, which is the pressure where fully pressurized ammonia is transported and stored as a marine fuel [37]. The temperature of system inlet ammonia is set to 45°C, which is the saturation temperature of ammonia at 18 bar. The pressure difference between anode and cathode inlet at the SOFC stack is set to be under 10 mbar to avoid mechanical destruction of the cell [38]. The temperature of the stack inlet stream is set to be minimum at 600°C among the parametric study to guarantee complete decomposition of ammonia at the fuel reformer and stack [32]. In addition, the temperature difference between the inlet stream and outlet stream of SOFC stack is set as 100°C to reduce thermal stress to the stack [39]. The mass flow rate of air through cathode inlet was adjusted at each operating conditions to satisfy the temperature difference boundary condition, as air can be utilized as a coolant to avoid overheating inside the stack. As fluid flows across the stack, it is assumed that a pressure drop of 0.5% occurs due to the geometry of anode and cathode channels [15].

Table 1. Input parameters and boundary conditions throughout this study.

Parameter	Value	Unit	Ref.
System boundary conditions			
System inlet air temperature	25	°C	
System inlet air pressure	1	atm	
Chemical composition of air	21% O ₂ , 79% N ₂	%	
System inlet fuel temperature	45	°C	[37]
System inlet fuel pressure	18	bar	[37]
Chemical composition of fuel	100% NH ₃	%	
System sink pressure	1	atm	
System sink temperature	Dew point	°C	
SOFC stack			
Number of stacked cells	2380	EA	
Effective cell area	121	cm ²	
Anode pressure drop	0.5	%	[15]
Cathode pressure drop	0.5	%	[15]
SOFC inlet pressure difference tolerance	10	mbar	[38]
SOFC inlet/outlet temperature difference	100	°C	[39]
Blower			
Isentropic efficiency	85	%	

The operating conditions of the system analyzed in this study are varied in a wide range with a base case stack inlet temperature (T_{stack}) of 600 °C, stack fuel utilization rate 70% (FU_{stack}), current density 0.2 A/cm² (j), AOG ratio 50% (r), which is a condition to generate approximately 50 kW of gross power. Stack fuel utilization and AOG ratio are defined as Eqs. (4) and (5).

$$FU_{stack} = \frac{\dot{n}_{O^{2-}}}{\left(\dot{n}_{H_2} + \frac{3}{2}\dot{n}_{NH_3}\right)_{stack,in}} \quad (4)$$

$$r = \frac{\dot{n}_{AOG}}{\dot{n}_{stack,a,out}} \quad (5)$$

Where $\dot{n}_{O^{2-}}$ is the rate of oxygen ion transport, defined as Eq. (6).

$$\dot{n}_{O^{2-}} = \frac{jN_{cell}A_{cell}}{2F} \quad (6)$$

When AOG is applied, unused fuel from the SOFC stack outlet is recirculated, and the recirculated fuel supplies the

fuel demand at the anode inlet. Such process reduces the amount of fuel required at the system inlet while satisfying FU_{stack} conditions; hence the system fuel utilization is determined as Eq. (7).

$$FU_{system} = \frac{FU_{stack}}{1 - r(1 - FU_{stack})} \quad (7)$$

Accordingly, mass flow rates of fuel inlet at each operating condition is determined as Eq. (8).

$$\dot{m}_{NH_3} = \frac{2\dot{n}_{O_2} - MW_{NH_3}}{3FU_{system}} \quad (8)$$

Where MW_{NH_3} is the molar mass of ammonia. To meet the power demand under given operating conditions, the SOFC stack is consisted of 2380 cells with an active area of 121 cm².

Dimensional parameters for the heat exchanger and reformer were determined following standard design guidelines for commercial shell-and-tube heat exchangers [40, 41, 42]. Key geometric variables—including tube length, shell diameter, and baffle spacing—were tuned to ensure that the required heat duties are met over a wide operating range. To prevent overdesign, the final dimensions were selected such that the tube-side velocity remained within an appropriate range (15-30m/s) for effective heat transfer [43]. However, since fluid flow rates vary substantially across operating conditions, a single heat exchanger design cannot satisfy the heat duty and velocity requirements at every point, even with splitter fluid control upstream of the heat exchanger or reformer. Therefore, if a given heat exchanger unit could not physically meet its heat duty requirements or tube-side velocity range, its size would be adjusted to meet such conditions. Details about the adjustment of heat exchanger and reformer dimensions is described in Supplementary

Table 2. Heat exchanger & reformer design configurations for base case operation

Parameter	Value	Unit	Ref.
Air heat exchanger			
Shell diameter	0.643	m	[40]
Tube diameter	0.0127	m	
Number of tubes	945	EA	
Baffle spacing	0.214	m	
Ammonia reformer			
Shell diameter	0.308	m	[40]
Tube diameter	0.0127	m	
Number of tubes	134	EA	
Baffle spacing	0.103	m	
Catalyst pellet diameter	0.002974	m	
Catalyst pellet length	0.01041	m	

Table 3. Variations in the operating parameters.

Operating parameters	Values	Unit
T_{stack}	600, 650, 700, 750, 800	°C
FU_{stack}	40, 50, 60, 70, 80	%
j	0.1, 0.2, 0.3, 0.4, 0.5	A/cm ²
AOGR ratio	0, 10, 20, 30, 40, 50... Max	%

Note S1. The main geometric parameters of the base-case heat exchanger and reformer are summarized in Table 2.

3.3. Operating parameters for parametric study

To analyze the operating limits and performance effects of AOGR ratio via parametric study, the following operating parameters were varied together with AOGR ratio: T_{stack} , FU_{stack} , j . The range and values of parameter variation are described in Table 3. Physically attainable values of maximum AOGR ratio are explained in detail in Section 4.2.

4. Results and Discussion

To analyze the sole effect of each operating parameter, each parameter was primarily varied at the base case. The complete stream table for the base case operation is provided in Supplementary Table S2. Key performance indices include system gross power, electrical efficiency (η_{elec}), etc. In this study, the only parasitic power induced in the system is power consumption at the air blower ($\dot{W}_{blower}$). Therefore,

Table 4. Simulated performance indices of the base case operation.

Parameter	Value	Unit
Unit cell voltage	0.864	V
Gross power	49.74	kW
Power density	0.173	W/cm ²
Fuel thermal input (LHV)	76.54	kW
Air blower power	1.35	kW
Net power	48.40	kW
System fuel utilization	82.4	%
Electrical efficiency	63.23	%

electrical efficiency and net power ($\dot{W}_{net}$) can be defined as Eqs. (9) and (10). To evaluate the performance of the system's capability to generate electrical power at a non-biased scale, the gross power was normalized as power density ($\dot{w}_{gross}$), defined as Eq. (11).

$$\eta_{elec} = \frac{\dot{W}_{net}}{\dot{n}_{fuel} LHV_{fuel}} \times 100, \quad (9)$$

$$\dot{W}_{net} = \dot{W}_{gross} - \dot{W}_{blower}, \quad (10)$$

$$\dot{w}_{gross} = \frac{\dot{W}_{gross}}{N_{cell} A_{cell}}. \quad (11)$$

4.1. Base-case system performance

Prior to the parametric investigation, simulation results of the base case (600 °C T_{stack} , 70% FU_{stack} , 0.2 A/cm² j , 50% r) are examined to establish a quantitative reference against which the trends discussed in the subsequent sections can be interpreted. The performance indices obtained at the base case are summarized in Table 4. The stack operates at a unit cell voltage of 0.864 V, which yields a gross power of 49.74 kW and a power density of 0.173 W/cm². Against a fuel thermal input of 76.54 kW on an LHV basis, the system consumes 1.35 kW of parasitic power at the air blower, resulting in a net power of 48.40 kW and an electrical efficiency of 63.23%. Since half of the unreacted fuel leaving the anode is returned to the ejector, the FU_{system} defined in Eq. (7) reaches 82.4%, appreciably higher than the 70% imposed at the stack.

4.2. Determination of maximum AOGR Ratio

Eq. (7) shows that greater r attains greater FU_{system} ; such trend raises the question of how far the AOGR ratio can actually be increased, as greater r and FU_{system} can potentially improve electrical efficiency by reducing fuel consumption. Determining this limit as a function of the operating condition is accordingly a prerequisite for the parametric analysis of Section 4.3, in which the AOGR ratio is swept up to its maximum allowable value at every operating point.

To inspect in detail the mechanism of determination of maximum AOGR ratio, the streams of the anode side gas and its recirculated portion can be detached from the overall system to compose an AOGR loop section as Fig. 2.

As a mixture of system inlet fuel and the recirculated portion of anode off-gas (stream 2) flows consecutively through the fuel reformer and stack anode channel, the stream undergoes pressure drop and exits the stack (stream 4). A portion of the low-pressure anode off-gas is branched off through the splitter depending on the AOGR ratio (stream 5) and enters the suction inlet of the ejector. The pressure of stream 5 must be elevated to compensate for pressure drop effects of the fuel reformer and stack anode channel. Therefore, a high-pressure fuel (stream 1) enters through the primary inlet of the ejector and is mixed with stream 5 to pressurize the recirculated fluid. The mixed fluid leaves the ejector at a back pressure as stream 2, and the pressure balance of AOGR loop is satisfied.

As the system AOGR ratio increases, the ejector entrainment ratio also increases due to the greater mass flow rate of stream 5. Under this condition, a larger mass flow rate of the low-pressure fluid in stream 5 is entrained into the ejector, and therefore the pressure of stream 1 must be increased to ensure proper mixing with stream 5 and to achieve the

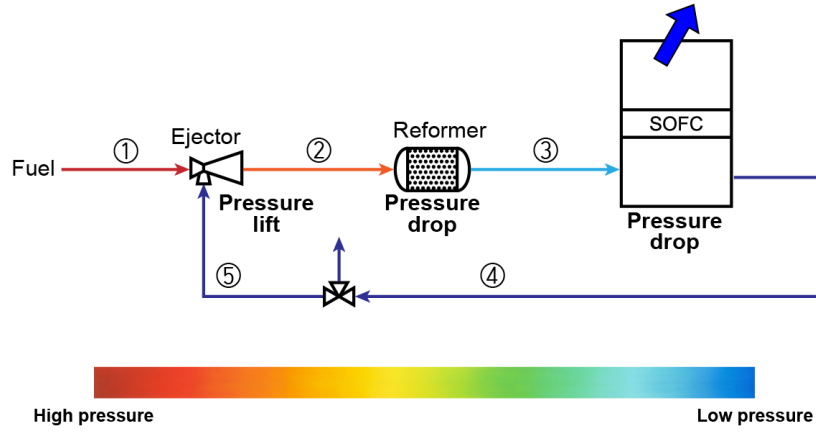


Figure 2: Schematic diagram of the AOGR loop section and pressure distribution in the constituent streams.

desired back pressure at stream 2. This is because a higher primary inlet pressure increases the stagnation pressure of the primary flow, thereby supplying more kinetic energy to the mixed fluid and resulting in greater pressure recovery [44]. However, the maximum pressure of stream 1 is limited to 18 bar, and the 18 bar primary flow may not be able to sufficiently pressurize the recirculated stream when the AOGR ratio is too high. In addition to this, the pressure drop occurring at the fuel reformer cold side also generally increases with the AOGR ratio due to an increase in the mass flow rate of stream 2. The maximum feasible AOGR ratio corresponds to the point at which the ejector, even at the maximum primary inlet pressure of 18 bar, can no longer provide sufficient pressure lift to overcome the pressure drop along the recirculation path and satisfy the required pressure condition at stream 2. Therefore, it can be deduced that the maximum AOGR ratio of the system is determined by the following two factors: entrainment and pressure lift performance of the ejector, and pressure drop occurring at the fuel reformer cold side.

Changes in operating conditions of the SOFC system (FU_{stack} , j , T_{stack}) can affect both the performance of the ejector and fuel reformer cold side pressure drop. Therefore,

it is necessary to analyze how a change in each operating condition affects the maximum AOGR ratio permitted to the system. Corresponding trends in maximum AOGR ratio with operating conditions is shown in Fig. 3.

4.2.1. Effect of stack fuel utilization on maximum AOGR ratio

Increasing FU_{stack} from 40% to 80% expands the feasible AOGR operating window, raising the maximum allowable AOGR ratio from 65.4% to 66.9% as shown in Fig. 3(a). This occurs because the ammonia fuel flow rate supplied to the system inlet is inversely proportional to FU_{system} by Eq. (8), and a higher FU_{stack} raises FU_{system} at any given AOGR ratio, so that less fresh fuel is required to sustain the target electrochemical operating condition of the SOFC stack and the mass flow rate of the anode-side gases passing through the fuel reformer cold side is reduced accordingly. For a given packed bed reformer geometry, this reduction in flow rate diminishes the viscous friction effect within the catalyst bed. The reformer cold side pressure drop accordingly decreases monotonically from 8.76 kPa at 40% FU_{stack} to 6.66 kPa at 80% FU_{stack} . The lower cold side pressure drop alleviates the discharge back pressure that the ejector must overcome in the AOGR loop, thereby relaxing

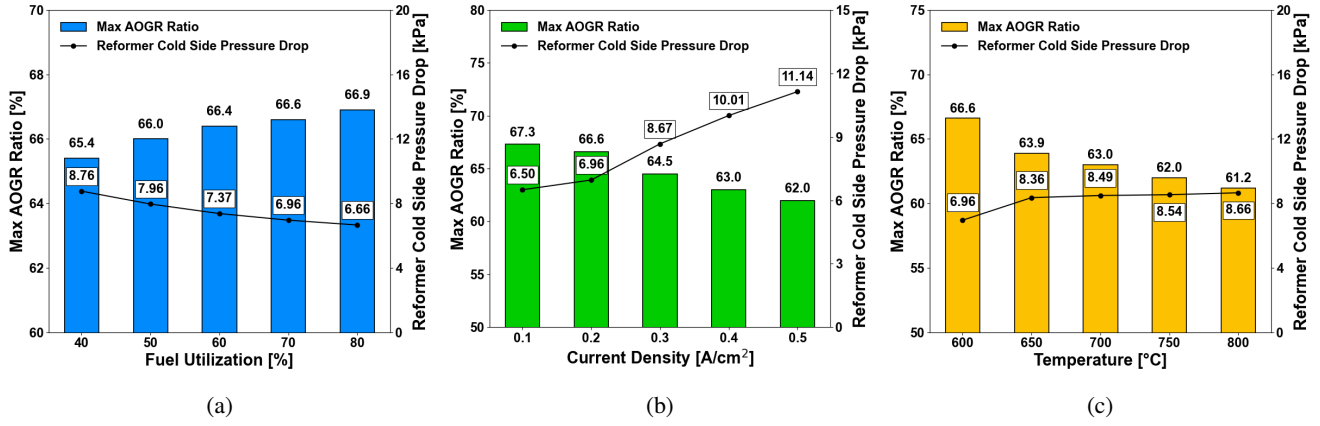


Figure 3: Maximum AOGR ratio permitted for variations in (a) FU_{stack} , (b) j , and (c) T_{stack} .

the pressure constraint that governs the upper bound of the AOGR ratio.

4.2.2. Effect of current density on maximum AOGR ratio

Increasing j from 0.1 A/cm^2 to 0.5 A/cm^2 acts in the opposite direction and narrows the feasible AOGR operating window, lowering the maximum allowable AOGR ratio from 67.3% to 62.0% as shown in Fig. 3(b). Two effects contribute to this reduction. First, j determines the amount of fuel electrochemically consumed in the SOFC stack, so that a higher j raises the ammonia fuel flow rate required at the system inlet and hence the mass flow rate of the anode-side gases passing through the fuel reformer cold side. Second, a higher j leads to a larger heat duty for the endothermic ammonia decomposition reaction, which requires a longer fuel reformer length to provide sufficient heat transfer area. As shown in Fig. 3(b), the combined effect of the increased flow rate and the extended reformer length amplifies the viscous friction loss along the packed catalyst bed, and the reformer cold side pressure drop increases monotonically from 6.50 kPa at 0.1 A/cm^2 to 11.14 kPa at 0.5 A/cm^2 . The higher cold side pressure drop intensifies the discharge back pressure imposed on the ejector, thereby tightening

the pressure constraint that governs the upper bound of the AOGR ratio.

4.2.3. Effect of stack inlet temperature on maximum AOGR ratio

Unlike FU_{stack} and j , T_{stack} restricts the AOGR operating window through a mechanism that the reformer cold side pressure drop alone cannot account for. Raising T_{stack} from 600 $^{\circ}C$ to 800 $^{\circ}C$ lowers the maximum allowable AOGR ratio from 66.6% to 61.2%, whereas the reformer cold side pressure drop at max AOGR ratio increases only from 6.96 kPa to 8.66 kPa over the same range, as shown in Fig. 3(c). The pressure drop increase itself is readily explained: T_{stack} determines the temperature of the gas entering the fuel reformer cold side, and the elevated gas temperature reduces the gas density at a fixed mass flow rate, which increases the volumetric flow velocity and hence the viscous friction loss along the packed catalyst bed. This modest increase, however, is insufficient to account for the magnitude of the reduction in the maximum allowable AOGR ratio. Since T_{stack} also determines the temperature of the entrained anode off-gas returning to the ejector, the remainder of the reduction must originate from the entrainment performance of the ejector itself.

Specifically, the deterioration originates from the manner in which the entrained stream temperature enters the one-dimensional description of the ejector, which follows the critical-mode formulation of Huang et al. [45]. Since the entrained stream constitutes the larger part of the mixture, its temperature largely governs the stagnation temperature—and hence the speed of sound—of the mixed stream, so that a hotter entrained flow reaches the shock at a lower Mach number. The static pressure rise across the shock scales with $M_m^2 - 1$, so that even a small reduction in the mixed flow Mach number, which remains only slightly supersonic, translates into a disproportionately large loss of pressure lift. To verify this, a supplementary analysis was conducted on a simplified double inlet-ejector-single sink system inhaling pure ammonia as primary flow and anode off-gas as entrained flow, both fixed at the base-case conditions, while the entrained flow temperature was varied from 700 °C to 900 °C. The results confirm that both the Mach number of the mixed flow and the attainable back pressure decrease monotonically as the entrained flow temperature increases (Supplementary Tables S3 and S4). Consequently, the ejector becomes less capable of overcoming the discharge back pressure of the AOGR loop, and the maximum allowable AOGR ratio decreases more sharply than the reformer pressure drop alone would predict.

4.3. Effect of AOGR ratio at different operating conditions

While the previous section identified the maximum allowable AOGR ratio under each operating condition, the AOGR ratio itself is a key design variable that strongly influences the overall system performance within the feasible operating window. In this section, the effect of the AOGR ratio on the system performance is investigated under varying operating conditions, namely FU_{stack} , j , and T_{stack} .

For each operating condition, the AOGR ratio is varied from zero up to the maximum allowable value determined in Section 4.2, and the resulting trends in electrical efficiency and key state variables are analyzed. This two-dimensional parametric analysis enables a comprehensive understanding of how the AOGR ratio interacts with each operating condition to shape the overall performance characteristics of the ammonia-fueled SOFC system. In each of these two-dimensional sweeps, only one of FU_{stack} , j and T_{stack} is varied together with the AOGR ratio, while the remaining two are held at their base-case values.

4.3.1. Effect of AOGR ratio at different stack fuel utilizations

To investigate the influence of the AOGR ratio on the system performance under varying FU_{stack} , the AOGR ratio was swept from 0% up to the maximum allowable value identified in Section 4.2, while FU_{stack} was varied from 40% to 80%. The resulting trends are summarized in Fig. 4: Fig. 4(a) presents electrical efficiency and power density, while Fig. 4(b) and Fig. 4(c) present system heat input and unit cell voltage, respectively. The trajectory of the system operating point as the AOGR ratio increases can be traced for each FU_{stack} .

As shown in Fig. 4(a), increasing the AOGR ratio consistently improves electrical efficiency while reducing stack power density across all FU_{stack} conditions; however, it can be seen that low FU_{stack} operation benefits more from AOGR compared to high FU_{stack} conditions. The efficiency gain is substantially greater at low FU_{stack} : at a FU_{stack} of 40%, raising the AOGR ratio from 0% to its maximum allowable value of 65.4% increases the electrical efficiency from 33.9% to 51.3%, an absolute gain of 17.4 percentage points. At a high FU_{stack} of 80%, the same operation only

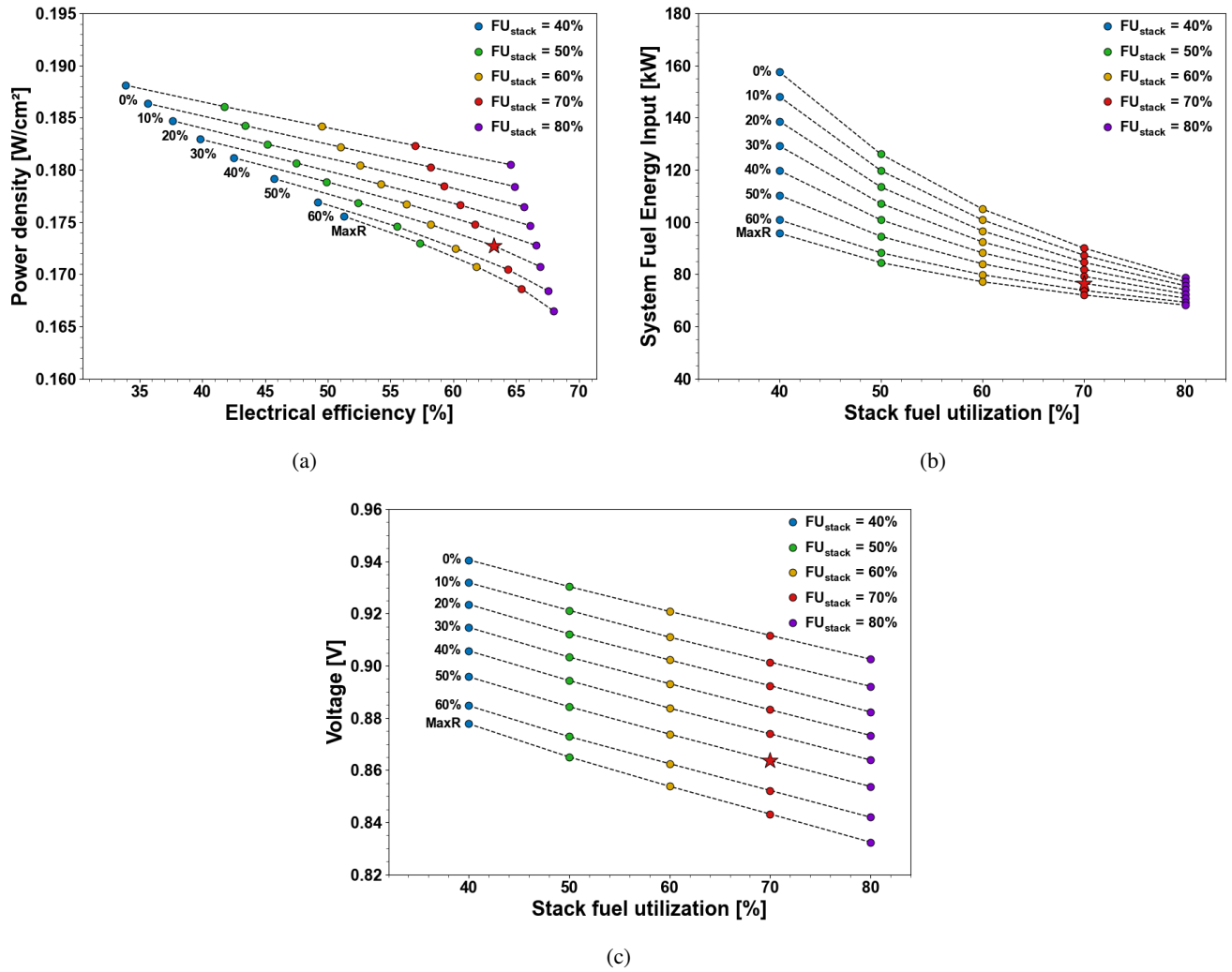


Figure 4: Performance map showing (a) electrical efficiency and power density, (b) system fuel energy input (LHV basis), and (c) voltage as a function of FU_{stack} and AOGR ratio. The blue, green, yellow, red, purple markers represent FU_{stack} condition of 40, 50, 60, 70, 80%, respectively. Number labels on the markers represent AOGR ratio, and the star marker represents base case operation.

improves the efficiency from 64.5% to 68.0%, yielding a much smaller gain of 3.5 percentage points.

The origin of this asymmetry can be explained by the AOGR-induced reduction in the system fuel input. As given by Eq. (8), the mass flow rate of ammonia required at the system inlet is inversely proportional to FU_{system} , so that the entire benefit of AOGR on the fuel side is determined by how far the AOGR ratio can lift FU_{system} above FU_{stack} . Since the fuel available for AOGR is the unutilized fraction $(1 - FU_{stack})$ exiting the stack, Eq. (7) indicates that this amplification progressively diminishes as FU_{stack} increases. At a FU_{stack} of 40%, raising the AOGR ratio to its maximum allowable value of 65.4% lifts FU_{system} from 40.0% to 65.8%, a 1.65-fold increase, whereas at a FU_{stack} of 80% the maximum allowable value of 66.9% lifts it only from 80.0% to 92.4%, a 1.16-fold increase. As shown in Fig. 4(b), the system fuel input expressed in terms of thermal energy follows these factors exactly, decreasing from 157.6 kW to 95.7 kW at $FU_{stack} = 40\%$ but only from 78.8 kW to 68.2 kW at $FU_{stack} = 80\%$ —a disparity that directly translates into the efficiency gain disparity observed in Fig. 4(a).

The stack power density penalty associated with AOGR is also greater for high FU_{stack} operation, but is relatively minor compared to the efficiency penalty. As the AOGR ratio increases, the recirculated stream—rich in N_2 and H_2O —dilutes the fresh decomposed fuel and lowers the hydrogen partial pressure at the stack anode inlet. According to the Nernst equation, this reduction suppresses the reversible potential and consequently the cell voltage. As shown in Fig. 4(c), increasing the AOGR ratio from 0% to its maximum value reduces the cell voltage from 0.941 V to 0.878 V at $FU_{stack} = 40\%$ (a drop of 0.063 V), and from 0.903 V to 0.832 V at $FU_{stack} = 80\%$ (a drop of 0.071 V). This behavior arises because AOGR causes greater fuel dilution at high FU_{stack} , where the anode off-gas already has a low fuel concentration. Consistent with this, the resulting power density reductions—from 0.188 W/cm² to 0.176 W/cm² at $FU_{stack} = 40\%$ and from 0.181 W/cm² to 0.166 W/cm² at $FU_{stack} = 80\%$ —are greater at $FU = 80\%$ operation, but differ by only 0.003 W/cm². Such results show that the efficiency benefit obtainable from AOGR is greatest at low FU_{stack} .

4.3.2. Effect of AOGR ratio at different current density

System performance was next examined under varying j and AOGR ratio, and the results are presented in Fig. 5. Fig. 5(a) shows the trend in efficiency and power density, and Fig. 5(b) shows the unit cell voltage. As shown in Fig. 5(a) and its inset plot, a high j does not merely weaken the efficiency benefit of AOGR but reverses it beyond a certain AOGR ratio: at 0.5 A/cm², the electrical efficiency increases with the AOGR ratio only up to approximately 60%, and then decreases as the AOGR ratio is further raised toward its maximum allowable value. The weakening of the benefit itself is evident from the two extremes of the j range. At a low j of 0.1 A/cm², increasing the AOGR ratio from 0% to its

maximum allowable value of 67.3% improves the electrical efficiency from 60.7% to 69.9%, corresponding to an absolute gain of 9.2 percentage points. At a high j of 0.5 A/cm², however, the same operation only enhances the efficiency from 46.3% to 49.8%, yielding a much smaller gain of 3.5 percentage points. The existence of an internal optimum at high j indicates that AOGR introduces a competing penalty of parasitic power consumption that becomes increasingly dominant as j increases.

In contrast, the trend in stack power density with increasing AOGR ratio is less sensitive to j . The relative reductions in power density—from 0.096 W/cm² to 0.089 W/cm² at 0.1 A/cm² and from 0.395 W/cm² to 0.366 W/cm² at 0.5 A/cm²—correspond to nearly the same value of approximately 7% in both cases. Such trend is followed by the trend in cell voltage, where as shown in Fig. 5(b), the cell voltage at any given j decreases by a comparable magnitude as the AOGR ratio is increased, regardless of the absolute j level. Specifically, increasing the AOGR ratio from 0% to its maximum value reduces the cell voltage from 0.962 V to 0.890 V at 0.1 A/cm² (a drop of 0.072 V) and from 0.790 V to 0.732 V at 0.5 A/cm² (a drop of 0.058 V). The slightly larger voltage drop observed at low j is partly attributable to the significantly wider AOGR sweep range, since the maximum allowable AOGR ratio is 67.3% at 0.1 A/cm² but only 62.0% at 0.5 A/cm². Such trend in power density degradation across j conditions can be understood from the fact that j does not significantly alter the molar composition at the stack inlet and outlet. Consequently, the resulting reduction in the anode hydrogen partial pressure is governed primarily by the AOGR ratio itself rather than by the operating j . Since the power density penalty alone cannot account for the non-monotonic efficiency behavior observed

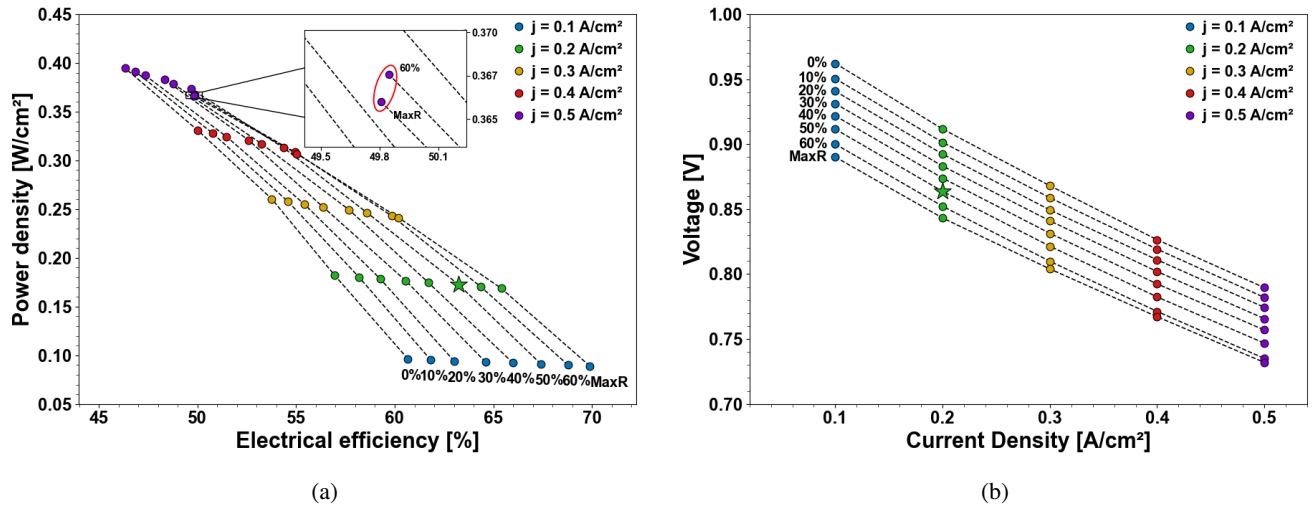


Figure 5: Performance map showing (a) electrical efficiency and power density and (b) voltage as a function of j and AORR ratio. The blue, green, yellow, red, purple markers represent j condition of 0.1, 0.2, 0.3, 0.4, 0.5 A/cm², respectively. Number labels on the markers represent AORR ratio, and the star marker represents base case operation.

at high j , the source of the reversal must instead lie in the parasitic power consumption of the system BoP.

To clarify this, the system parasitic power was taken into analysis. As addressed in Eq. (10), the air blower is responsible for parasitic power consumption in the present system, and the pressure rise that the air blower must provide is determined by the cumulative pressure drop along the cathode-side flow path, given by Eq. (12).

$$p_{air\ blower} = 1\ atm + \Delta p_{stack} + \Delta p_{reformer,hot} + \Delta p_{air\ HX,hot} \quad (12)$$

where Δp_{stack} is the pressure drop across the SOFC stack cathode channel, $\Delta p_{reformer,hot}$ is the pressure drop across the hot side of the fuel reformer, and $\Delta p_{air\ HX,hot}$ is the pressure drop across the hot side of the air heat exchanger. The flow path of fluid to be pressurized by the air blower is represented in [Supplementary Fig. S2](#). As AORR ratio increases, the pressure drops occurring at the heat exchanger and fuel reformer are amplified, while the SOFC stack channel pressure drop is determined almost constant as 0.5% of the fluid pressure. First, the pressure drop

on the hot side of the fuel reformer rises through a more subtle mechanism than a simple increase in the cold-side flow rate. With increasing AORR ratio, the mass flow rate of fresh ammonia entering the reformer cold inlet decreases, since a larger fraction of the stack inlet fuel demand is met by recirculated unreacted fuel. This reduction in ammonia content moderately lowers the heat duty required for the endothermic ammonia decomposition reaction. However, the AORR ratio simultaneously raises the temperature of the reformer cold inlet stream, because the recirculated anode off-gas—returning at a high temperature—constitutes an increasingly large fraction of the cold stream. Since the reformer hot stream inlet temperature is essentially fixed as the stack cathode off gas temperature, the elevated cold-side temperature causes the log-mean temperature difference (LMTD) across the reformer to decrease substantially. Because the LMTD reduction outweighs the moderate decrease in the heat duty, the mass flow rate of the hot stream required to satisfy the heat balance must increase, leading to a higher pressure drop $\Delta p_{reformer,hot}$ on the hot side of the fuel reformer. The corresponding data of LMTD and heat duty at the fuel reformer is provided in [Supplementary Table S5](#) for

Table 5. Summary of parasitic power and pressure drops for different AOGR ratios at different j values

Parameters	Values										Units
j	0.1		0.2		0.3		0.4		0.5		A/cm ²
AOGR ratio	0	Max	0	Max	0	Max	0	Max	0	Max	%
$\Delta p_{reformer,hot}$	1.83	3.20	2.50	3.89	3.21	6.01	3.32	8.01	4.15	9.93	kPa
$\Delta p_{air\ HX,hot}$	1.92	3.04	3.49	3.69	5.23	6.69	8.32	10.37	12.04	14.68	kPa
$\dot{W}_{blower}$	0.31	0.52	1.23	1.44	2.37	3.88	5.11	7.99	9.42	14.13	kW

different AOGR ratio operation. Second, the pressure drop on the hot side of the air heat exchanger rises through a chain of effects originating from the boundary condition that the temperature difference between the stack inlet and outlet must remain at 100°C, which determines the cathode air flow rate required for cooling. As discussed earlier, increasing the AOGR ratio reduces the cell voltage and hence the stack power generation; since the chemical energy released by the electrochemical reaction is partitioned between electrical work and heat, the heat to be removed from the stack increases accordingly, and the cathode air flow rate must be raised to dissipate it. The larger air flow rate elevates the heat capacity flow rate ($\dot{m}C_p$) on the cold side of the air heat exchanger, so that the mass flow rate of the hot stream supplying it must also increase to satisfy the heat balance and $\Delta p_{air\ HX,hot}$ grows with the AOGR ratio. Combined with the direct increase in air flow rate through the blower itself, these compounding effects cause the air blower parasitic power to grow rapidly with the AOGR ratio.

The relative impact of this parasitic power growth on the overall electrical efficiency depends strongly on the operating j , since the absolute magnitude of both air flow rate and pressure drops occurring at the heat exchanger and fuel reformer scales with j . As summarized in Table 5, at a low j of 0.1 A/cm², the air blower parasitic power increases from 0.31 kW at zero recirculation to 0.52 kW at the maximum AOGR ratio.

This increase is accompanied by a moderate rise in both hot side pressure drops, as listed in Table 5. Although the absolute increase in parasitic power (≈ 210 W) is non-negligible, it remains a minor fraction of the gross stack power output, corresponding to approximately 1% of the gross power generated under this operating condition. As a result, at low j the efficiency gain from reduced fuel input clearly dominates over the parasitic penalty, and the efficiency increases monotonically with the AOGR ratio. At a high j of 0.5 A/cm², in contrast, the air blower parasitic power soars from 9.42 kW at zero recirculation to 14.13 kW at the maximum AOGR ratio—an absolute increase of approximately 22 times larger than the low- j case. The associated hot side pressure drops also rise much more steeply. The increase in parasitic power at this condition reaches approximately 4.5% of the gross stack power output at the maximum AOGR ratio, which is significant enough to overcome the benefit of reduced fuel input beyond a certain AOGR ratio. Consequently, at 0.5 A/cm², the electrical efficiency continues to improve only up to about 60% AOGR; beyond this point, the marginal increase in parasitic power exceeds the marginal reduction in fuel input, and the efficiency begins to decline despite the continued recovery of unutilized fuel. As a result, the dominant trade-off under varying j shifts from the stack level to the balance-of-plant level, such that the optimal AOGR ratio at high j lies below

the maximum allowable value and decreases further as j_{19} increases.

4.3.3. Effect of AOGR ratio at different stack inlet temperatures

The effect of AOGR ratio on the system performance was finally examined under varying T_{stack} conditions, ranging from 600 °C to 800 °C. The resulting trajectories of electrical efficiency and stack power density are presented in Fig. 6(a), and unit cell voltage values are presented in Fig. 6(b). The central outcome from Fig. 6(a) is that the most efficient operating temperature is not fixed, but shifts downward as the AOGR ratio increases. At zero recirculation, the highest system performance is achieved at a T_{stack} of 750 °C, with the operating point at 700 °C closely following and effectively overlapping with the 750 °C case in the efficiency–power density plane at virtually identical values of 0.1908 W/cm² and 59.92%. The 600 °C and 650 °C operating points yield progressively lower performance than the 700–750 °C cases, while the 800 °C case—despite being the highest temperature considered—delivers performance that is consistently lower than that of the 700–750 °C cases throughout the entire AOGR range. This indicates that the cell voltage at zero recirculation already peaks within the 700–750 °C interval and operating at 800 °C lies on the descending side of the voltage–temperature curve. As the AOGR ratio increases, however, the trajectories of the higher-temperature cases (750 °C and 800 °C) curve downward more steeply than that of the 700 °C case, and the relative ranking among the operating temperatures evolves accordingly. At maximum allowable AOGR ratio, 700 °C emerges as the most efficient operating temperature with an electrical efficiency of 67.52%, while 750 °C falls slightly behind at 67.02% and 800 °C reaches only 66.02%.

This downward shift of the optimal operating temperature originates from the way the AOGR ratio modifies the dependence of the stack voltage on T_{stack} . As shown in Fig. 6(b), the cell voltage at any given AOGR ratio exhibits a non-monotonic dependence on T_{stack} : it first increases from 600°C to an intermediate maximum and then decreases again toward 800°C. An intermediate optimal temperature exists due to a trade-off in reversible potential and overpotential. As the operating temperature of the SOFC stack increases, its activation and ohmic overpotential decreases due to an increase in exchange current density and ionic conductivity, while the reversible potential decreases. Detailed information in stack voltage at different T_{stack} with base case $FU_{stack,j}$, AOGR ratio, is provided in [Supplementary Table S6](#). The location of this maximum, however, shifts to lower temperatures as the AOGR ratio increases. At zero recirculation, the cell voltage peaks broadly across the 700–750 °C range, with the two temperatures yielding nearly identical voltage values of 0.9541 V and 0.9542 V. As the AOGR ratio increases, the peak gradually narrows and migrates toward 700 °C: at the maximum allowable AOGR ratio, the cell voltage at 700 °C (0.8807 V) clearly exceeds that at 750 °C (0.8779 V), even though the two values were essentially indistinguishable at zero recirculation. Reflecting this temperature-dependent penalty, the cumulative voltage drops incurred over the full AOGR sweep increases monotonically with T_{stack} , from 68.5 mV at 600 °C to 73.3 mV at 700 °C, 76.3 mV at 750 °C, and 79.1 mV at 800 °C, even though the maximum available AOGR ratio is lower at a high T_{stack} . This trend confirms that AOGR introduces a temperature-dependent penalty mechanism whose magnitude grows with the AOGR ratio and disproportionately affects the higher-temperature operating points.

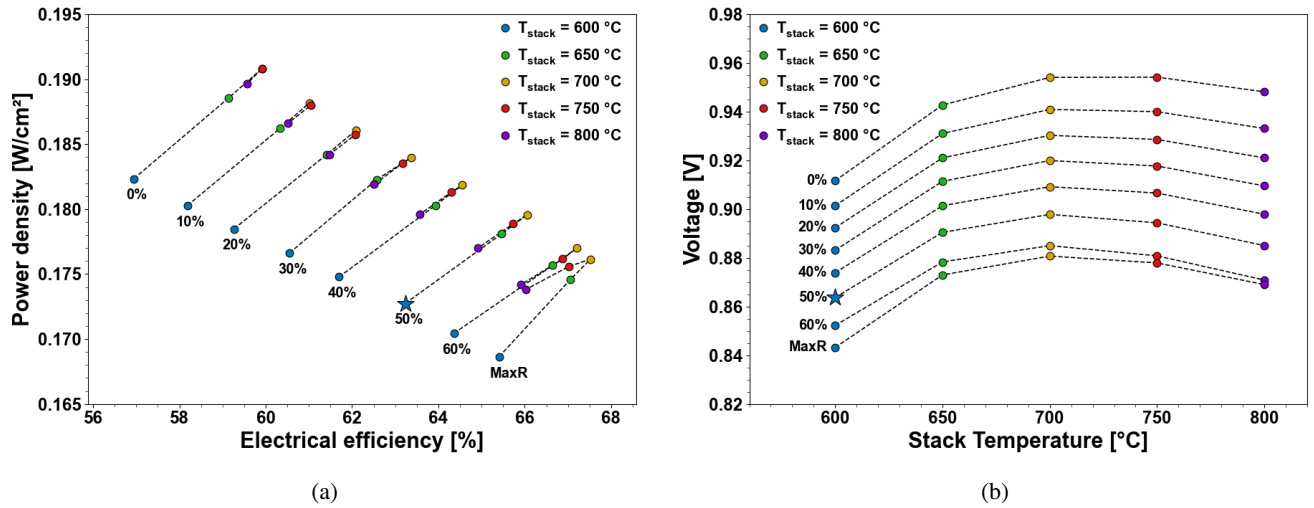


Figure 6: Performance map showing (a) electrical efficiency and power density and (b) voltage as a function of T_{stack} and AOGR ratio. The blue, green, yellow, red, purple markers represent T_{stack} condition of 600, 650, 700, 750, 800 °C, respectively. Number labels on the markers represent AOGR ratio, and the star marker represents base case operation.

The origin of this behavior can be explained by the dependence of reversible potential—which is the dominant factor determining cell voltage—on temperature and anode gas compositions, as higher temperatures reduce the reversible potential. The reversible cell voltage of the SOFC is given by Eq. (2), where the second term on the right-hand side—the Nernstian term—governs how this temperature dependence is modified by the anode gas composition, while the standard-potential term $-\Delta G^\circ(T)/2F$ sets its baseline magnitude. As the AOGR ratio increases, the recirculated stream depletes hydrogen partial pressure p_{H_2} and enriches the steam partial pressure p_{H_2O} at the stack anode inlet, which reduces the partial pressure ratio inside the logarithm to a value below unity and renders the logarithmic term increasingly negative. Therefore, at higher AOGR ratios, a given temperature increment produces a larger reduction in the Nernstian term and therefore a steeper decline in the reversible potential. To quantify the amplification of the reversible potential-reduction effect with AOGR, the magnitude of the reversible potential temperature sensitivity, $|dE_{rev}/dT|$, is shown in [Supplementary Fig. S3](#). This finding suggests that the design and operation of ammonia-fueled SOFC systems incorporating AOGR should explicitly account for the coupling between the AOGR ratio and T_{stack} , rather than treating the two parameters as independently optimizable variables. In particular, when high AOGR ratios are desired for electrical efficiency reasons, operating the stack at moderately high temperatures yields superior overall performance compared to operating at the upper end of the temperature range.

5. Conclusions

A comprehensive parametric investigation of an ammonia-fueled SOFC system incorporating anode off-gas recirculation (AOGR) was conducted to clarify both the operating limits imposed on the AOGR ratio and the influence of the AOGR ratio on the overall system performance. An integrated thermodynamic system model based on validated component models was constructed, and three operating parameters (FU_{stack} , j , and T_{stack}) were systematically varied together with the AOGR ratio. The principal findings are summarized as follows.

• The maximum allowable AOGR ratio is governed by the pressure constraint of the AOGR loop, in which the ejector pressure lift must overcome the cumulative pressure drops along the recirculation path. Increasing FU_{stack} expands this limit modestly by reducing the reformer cold-side pressure drop, whereas increasing j narrows it substantially due to the larger anode-side flow rate and the longer reformer length required to meet the elevated heat-duty demand.

• T_{stack} limits the maximum allowable AOGR ratio through a fundamentally distinct mechanism. The elevated entrained flow temperature degrades the pressure lift performance of the ejector, thereby reducing the maximum allowable AOGR ratio more sharply than what the pressure drop alone would predict.

• Under varying FU_{stack} , the AOGR ratio induces an asymmetric trade-off: the efficiency gain is highly sensitive to FU_{stack} , whereas the power density penalty is relatively insensitive. AOGR is therefore most advantageous at low FU_{stack} , where a substantial efficiency improvement is achieved at a similar power density cost as at high FU_{stack} .

• Under varying j , the dominant trade-off shifts from the stack level to the balance-of-plant level, giving rise to a non-monotonic efficiency trajectory at high j . The air blower parasitic power, amplified by the elevated air flow rate and the increased BoP pressure drops, eventually overcomes the benefit of reduced fuel input. Consequently, the optimal AOGR ratio lies below the maximum allowable value identified from the pressure constraint.

• Under varying T_{stack} , the optimal operating temperature shifts downward as the AOGR ratio increases,

since the AOGR-induced enrichment of steam at the anode inlet amplifies the reversible potential's temperature sensitivity. The resulting steeper reversible potential decline at higher temperatures overwhelms the modest overpotential reduction available under low-current-density operation, so that operation at the upper end of the investigated temperature range no longer yields the best performance under high AOGR conditions.

These findings demonstrate that the AOGR ratio cannot be treated as an independently optimizable variable, and the performance maps derived in this work provide quantitative guidance for selecting it together with each operating parameter in ammonia-fueled SOFC systems for marine power generation.

Despite the comprehensive scope of the present analysis, several limitations should be acknowledged and addressed in future work. First, the geometrical design variables of the BoP components—including the shell-and-tube heat exchanger and the packed-bed ammonia reformer—were not individually optimized for each operating condition examined in this study, although their dimensions were resized when the heat-duty requirements could not be satisfied. A condition-specific optimization of these geometrical variables could further refine the trade-offs identified in this work and yield more accurate quantification of the AOGR-induced parasitic power growth. Second, the present analysis was based on a single fixed system layout—the modified Ejector CG configuration—and alternative system architectures such as blower-driven recirculation, hybrid ejector-blower configurations, or layouts with different heat integration schemes were not considered, even though such variations may shift the location of the dominant trade-off mechanism and the resulting optimal AOGR ratio. Third, stack

durability and material degradation under prolonged AOGR operation were not included in the present scope, although the AOGR-induced enrichment of steam at the anode inlet identified in this work suggests that long-term durability considerations could further constrain the practical AOGR operating window beyond the steady-state thermodynamic limits established here. Future work should therefore address¹⁰⁸⁰ these aspects to translate the system-level design insights of this study into long-term operating guidelines.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have¹⁰⁸¹ appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

To be completed by the authors before submission.

Nomenclature

Latin symbols

A_{cell}	Effective cell area	cm ²
C_p	Specific heat capacity at constant pressure	kJ/(kg K)
E_{rev}	Reversible potential	V
F	Faraday constant	C/mol
FU_{stack}	Stack fuel utilization	%
FU_{system}	System fuel utilization	%
j	Current density	A/cm ²
LHV	Lower heating value	kJ/mol

$\dot{m}$	Mass flow rate	kg/h
MW	Molecular weight	g/mol
$\dot{n}$	Molar flow rate	mol/s
N_{cell}	Number of cells in the stack	—
p	Pressure	atm
r	AOGR ratio	%
R	Universal gas constant	J/(mol K)
T	Temperature	°C
T_{stack}	Stack inlet temperature	°C
V	Cell voltage	V
$\dot{W}$	Power	kW
$\dot{w}_{gross}$	Gross power density	W/cm ²
y_i	Mole fraction of species i	—

Greek symbols

ΔG°	Standard Gibbs free energy change of reaction	J/mol
Δp	Pressure drop	kPa
η_{act}	Activation overpotential	V
η_{conc}	Concentration overpotential	V
η_{elec}	System electrical efficiency	%
η_{ohm}	Ohmic overpotential	V

Subscripts and superscripts

a	Anode
act	Activation
$AOGR$	Recirculated anode off-gas
$air\ HX$	Air heat exchanger
$blower$	Air blower
c	Cathode
$cell$	Single cell
$conc$	Concentration
$elec$	Electrical
$fuel$	System inlet fuel

<i>gross</i>	Gross value	1105
<i>hot</i>	Hot side of a heat exchanger	1106
<i>i</i>	Chemical species	1107
<i>in</i>	Inlet	1108
<i>net</i>	Net value	1109
1085 <i>ohm</i>	Ohmic	1110
<i>out</i>	Outlet	1111
<i>reformer</i>	Ammonia reformer	1112
<i>rev</i>	Reversible	1113
<i>stack</i>	SOFC stack	1114
<i>system</i>	Overall system	1115
1086 Abbreviations		
AOGR	Anode off-gas recirculation	1116
BoP	Balance of plant	1117
1087 HX	Heat exchanger	1118
LMTD	Log-mean temperature difference	1119
SOFC	Solid oxide fuel cell	1120

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